

# Noise-Aware Transformer-Based Low-Light Image Enhancement with Downstream Object Detection Evaluation

Mid-semester B.Tech Project evaluation

Dhirendra Pratap Singh (2301EE49)  
Supervisor: Dr. Rajib Kumar Jha  
Department of Electrical Engineering, IIT Patna  
[MISSING: Date]

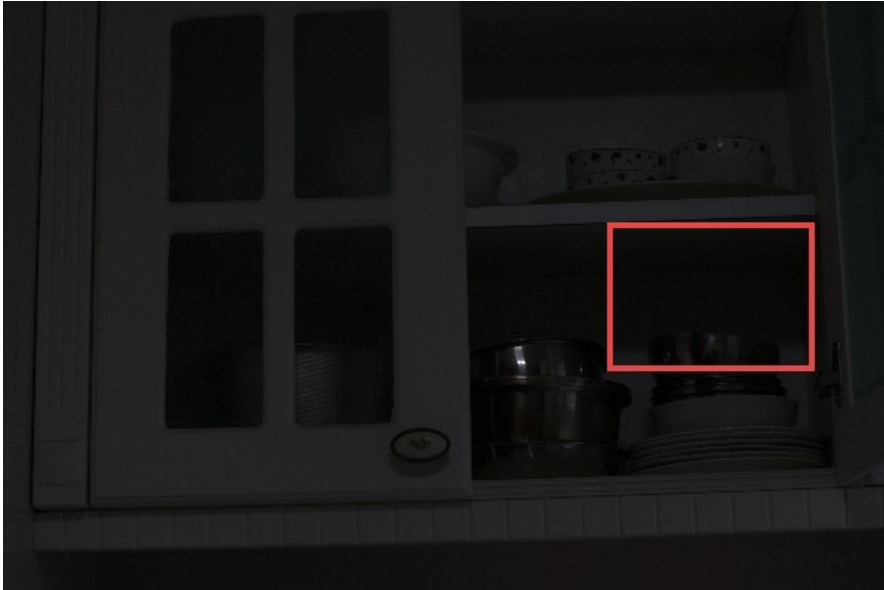
# Why low-light enhancement?



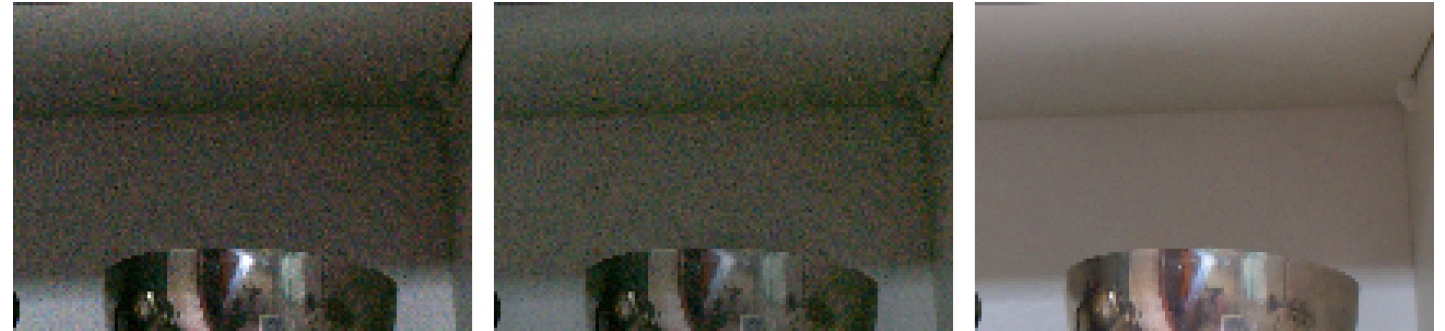
LOL-v1 test image 79: dark input (left) and Retinexformer output (right)

- Phone night photography
- Surveillance and security cameras
- Driving at night
- Machine vision: detectors fail in the dark
- Goal: bright, natural colours, no amplified noise

# The problem: darkness + noise + colour shift



Red box: dark region (LOL-v1 79)



Same crop: input  $\times$  6 | Zero-DCE | ground truth

- Dark: little light reaches the sensor
- Noisy: few photons  $\rightarrow$  low signal-to-noise ratio (SNR)
- Colour shift / over-exposed lamps
- Brightening multiplies the noise as much as the signal

# How the field evolved



- Hand-made rules → learned from data → no pairs needed → long-range attention → generative
- Survey (Li et al., TPAMI 2022): no method wins everywhere; noise in dark regions remains open
- This project: Retinexformer (ICCV 2023) as the main model

# Key methods compared in this project

| Method             | Year / venue | Key idea                                  | Limitation                          |
|--------------------|--------------|-------------------------------------------|-------------------------------------|
| HE / CLAHE / Gamma | classical    | Re-map brightness                         | No noise model, colour shifts       |
| Zero-DCE           | CVPR 2020    | Learned brightness curves, no pairs       | Brightens noise                     |
| SCI                | CVPR 2022    | Self-calibrated illumination, ~300 params | Brightens noise                     |
| SNR-Aware          | CVPR 2022    | SNR map decides local vs. long-range      | Heavier; no Retinex model           |
| LLFormer           | AAAI 2023    | Axis-based attention for 4K/8K images     | 24.5 M params, slower               |
| Retinexformer      | ICCV 2023    | Illumination-guided channel attention     | Illumination-aware, not noise-aware |
| GSAD               | NeurIPS 2023 | Diffusion with structure regularization   | Slow; GT-brightness test trick      |

# Research gap and objectives of this midsem

## Gaps

- Noise amplification in dark regions is unsolved
- Retinexformer: illumination-aware, not noise-aware
- Methods judged by PSNR/SSIM, rarely by a machine task
- Diffusion: quality at a large speed cost

## Objectives (first half)

- Reproduce Retinexformer + 8 baselines
- One protocol: PSNR, SSIM, LPIPS, NIQE, dark regions, speed
- Detection study: does enhancement help YOLOv8?
- Noise-aware SNR-weighted loss, tested against a control
- Plan the second half from the evidence

# Background: Retinex theory and attention cost

## Retinex model

$$I = R \odot L$$

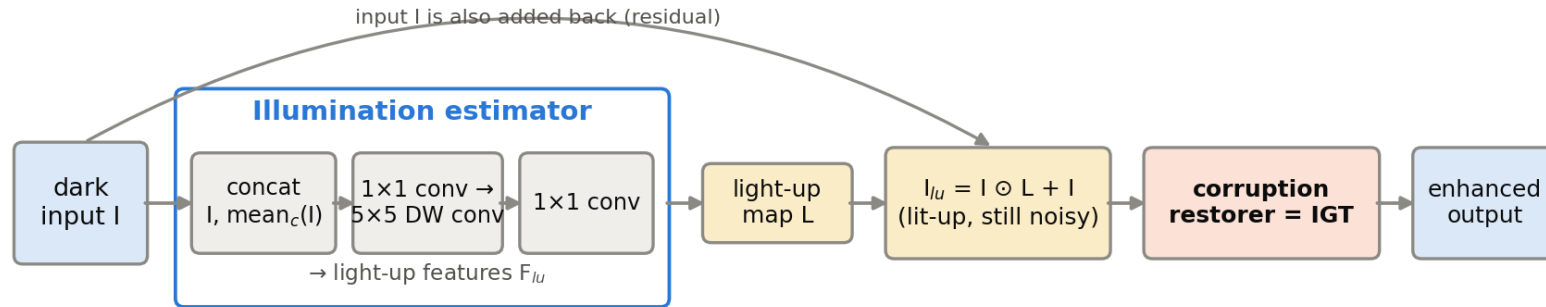
- $I$ : observed image
- $R$ : reflectance = object colours
- $L$ : illumination = light on the scene
- Enhance: estimate  $L$ , replace by brighter light
- Real images: noise in  $R$  and  $L$ , amplified when  $L$  is small

## Why attention is costly

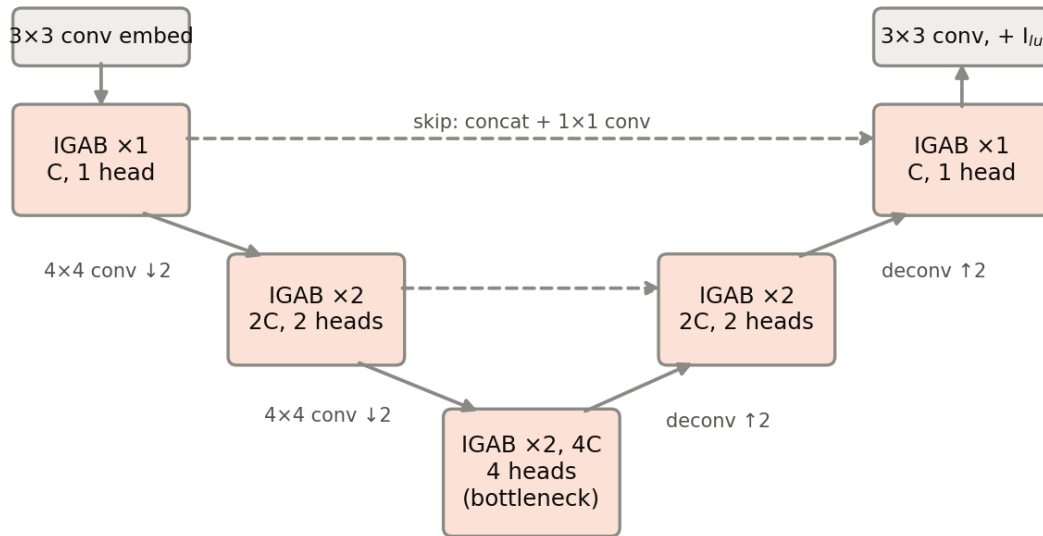
$$\text{softmax}(QK^T/\sqrt{d}) \cdot V$$

- $QK^T$  is  $N \times N$  ( $N$  = number of pixels)
- 600×400 image:  $N = 240,000 \rightarrow 58$  billion entries
- Channel attention:  $\text{softmax}(K^TQ)$  is  $C \times C$
- $\rightarrow$  cost grows linearly with image size

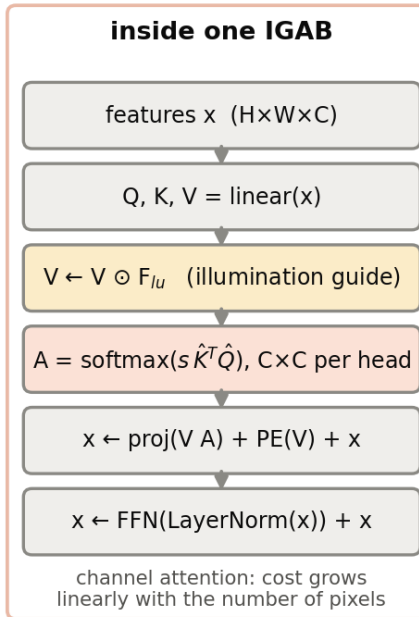
# Main model: Retinexformer (ICCV 2023)



## Illumination-Guided Transformer (IGT), C = 40 channels



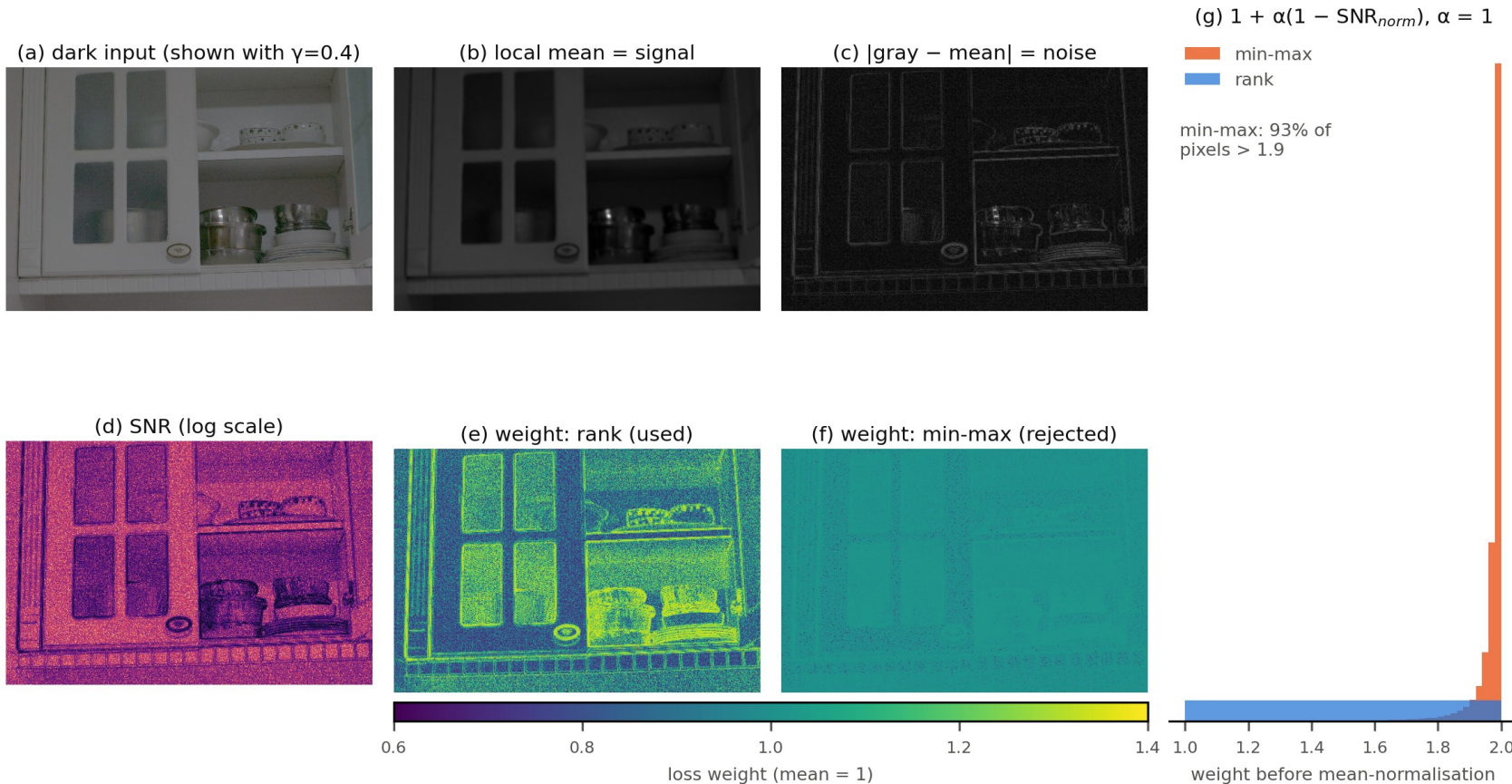
F<sub>Iu</sub> is down-sampled together with the features and fed to every IGAB.



- Estimator → light-up map L
- Lit-up image:  $I \odot L + I$  (still noisy)
- Restorer: U-shaped transformer (IGT)
- IG-MSA:  $V \times$  light-up features, attention across channels
- 1.61 M params,  $\ell_1$  loss



# My idea: noise-aware (SNR-weighted) loss



## Steps

- blur  $b = 5 \times 5$  mean of gray
- noise  $n = |\text{gray} - b|$
- $\text{SNR} = b / (n + \epsilon)$
- rank  $r \in [0,1]$  per image
- $w = 1 + \alpha(1 - r)$ ,  $\alpha = 1$
- $w \leftarrow w / \text{mean}(w)$
- $L = \text{mean}(w \cdot |\hat{Y} - Y|)$

Rank normalisation (e) spreads the weights; min-max (f) gives almost every pixel the same weight (93%  $> 1.9$ ).

# Experimental setup

| Data                | Use            | Test images          |
|---------------------|----------------|----------------------|
| LOL-v1              | paired, main   | 15                   |
| LOL-v2 real / syn   | paired         | 100 / 100            |
| LIME / DICM / MEF   | unpaired, NIQE | 10 / 69 / 17         |
| ExDark (12 classes) | detection      | 1,200 (200 for GSAD) |

| Metric        | Measures            | Better |
|---------------|---------------------|--------|
| PSNR          | pixel fidelity      | ↑      |
| SSIM          | structure           | ↑      |
| LPIPS         | perceptual distance | ↓      |
| NIQE          | naturalness (no GT) | ↓      |
| Dark-30% PSNR | darkest pixels only | ↑      |
| mAP50         | detection accuracy  | ↑      |

- Hardware: Kaggle Tesla T4 (training, timing); laptop for evaluation
- Rules: no GT at test time · no test-chosen checkpoint · no train/test overlap

# Quantitative results (paired test sets)

| Method               | LOL-v1 PSNR↑ | SSIM↑        | LPIPS↓       | LOL-v2-real PSNR↑ | SSIM↑        | LPIPS↓       |
|----------------------|--------------|--------------|--------------|-------------------|--------------|--------------|
| Input                | 7.77         | 0.191        | 0.560        | 9.72              | 0.196        | 0.519        |
| Gamma                | 14.53        | 0.620        | 0.351        | 18.70             | 0.659        | 0.318        |
| Zero-DCE             | 14.86        | 0.562        | 0.335        | 18.06             | 0.580        | 0.313        |
| SCI                  | 14.85        | 0.527        | 0.339        | 17.36             | 0.542        | 0.307        |
| SNR-Aware*           | 24.61        | 0.840        | 0.151        | 21.48             | <b>0.848</b> | 0.157        |
| LLFormer             | 23.65        | 0.816        | 0.169        | —                 | —            | —            |
| GSAD                 | 22.73        | <b>0.850</b> | <b>0.103</b> | 20.14             | 0.845        | <b>0.114</b> |
| Retinexformer        | <b>25.15</b> | 0.843        | 0.131        | <b>22.79</b>      | 0.839        | 0.171        |
| Retinexformer (mine) | 23.10        | 0.828        | 0.143        | —                 | —            | —            |

\* images released by the SNR-Aware authors. Best value in bold. —: train/test overlap (see setup).

# Reproduction and protocol findings

| Method / set               | Paper | Ours  |
|----------------------------|-------|-------|
| Retinexformer, LOL-v1      | 25.16 | 25.15 |
| SNR-Aware, LOL-v1          | 24.61 | 24.61 |
| LLFormer, LOL-v1           | 23.65 | 23.65 |
| GSAD, LOL-v1 (GT trick)    | 27.84 | 27.57 |
| GSAD, LOL-v1 (real output) | —     | 22.73 |

PSNR in dB on LOL-v1

- GSAD's GT-brightness trick: +4.85 dB on LOL-v1
- LOL-v2-real test: 91 / 100 images are LOL-v1 training images
- My re-training: 23.10 dB vs released 25.15 (others: 23.45)

# Qualitative results





# Does enhancement help a detector? (ExDark, YOLOv8m)

| Input to YOLOv8m     | mAP50 (1,200) | Recall       | mAP50 (200)  |
|----------------------|---------------|--------------|--------------|
| Raw (no enhancement) | <b>0.662</b>  | <b>0.611</b> | <b>0.707</b> |
| Retinexformer        | 0.603         | 0.540        | 0.628        |
| SCI                  | 0.595         | 0.529        | 0.625        |
| Zero-DCE             | 0.572         | 0.502        | 0.603        |
| GSAD                 | —             | —            | 0.633        |

- Raw images give the best mAP
- Enhancement lowers recall, not precision
- Better restoration hurts less, none helps



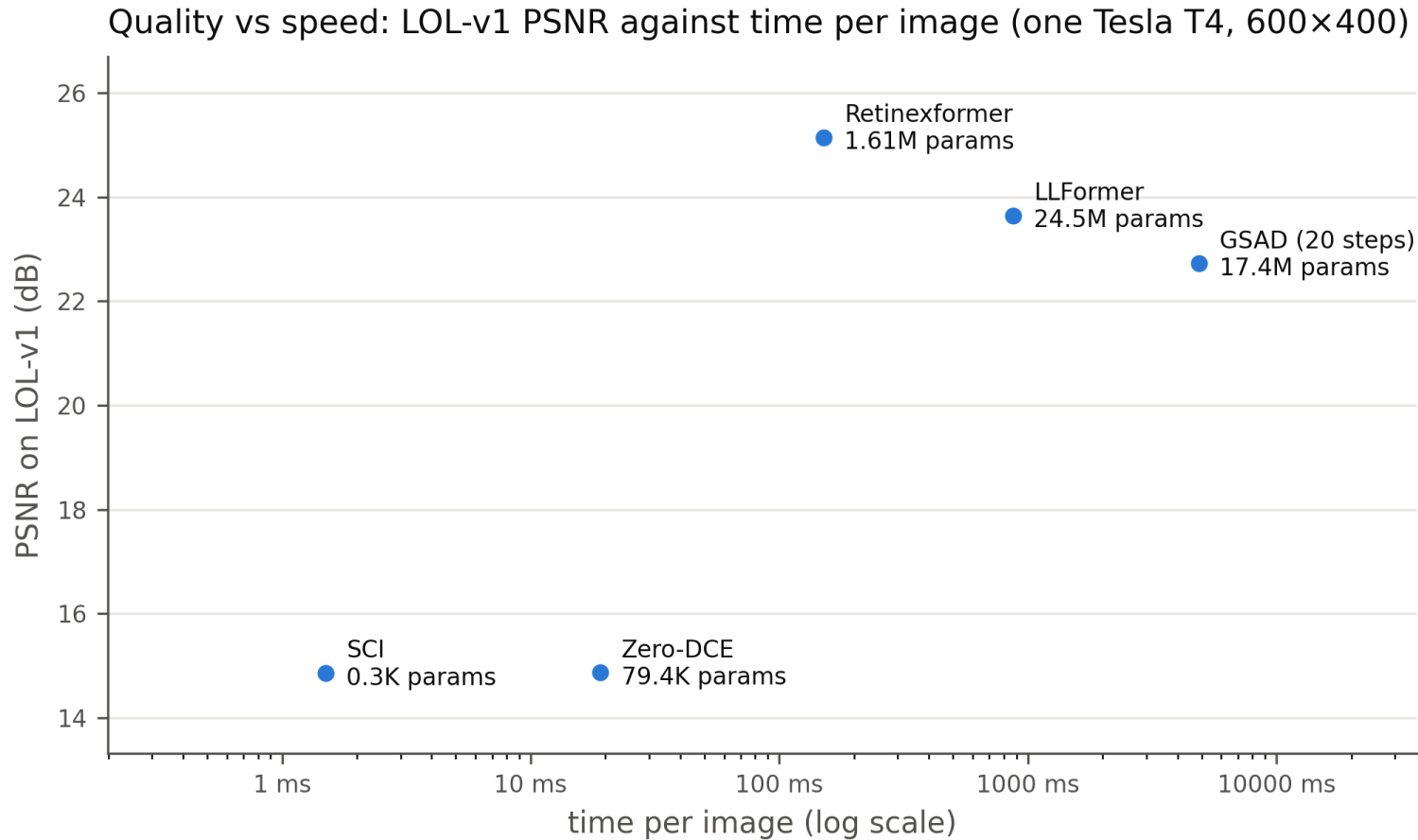
# Ablation: does the noise-aware loss help?

| Run              | LOL-v1 PSNR  | Dark-30% PSNR | Dark wins vs A | ExDark mAP50 |
|------------------|--------------|---------------|----------------|--------------|
| Released (start) | <b>25.15</b> | <b>25.31</b>  | –              | <b>0.627</b> |
| A: L1 (control)  | 24.31        | 24.89         | –              | <b>0.627</b> |
| B: SNR-L1 (mine) | 24.32        | 24.91         | 13/15          | 0.626        |
| C: L1 + FFT      | 24.31        | 24.89         | 3/15           | 0.626        |
| D: SNR-L1 + FFT  | 24.32        | 24.91         | 12/15          | 0.626        |

- B vs A: dark regions better on 13/15 images ( $p = 0.004$ )
- ...but only +0.01 to +0.05 dB: negligible
- FFT loss: no effect
- Fine-tuning lowers LOL-v1 PSNR for every run

**Takeaway: a consistent effect in the intended place (dark regions), but far too small to matter. The idea needs to act inside the network, not only in the loss.**

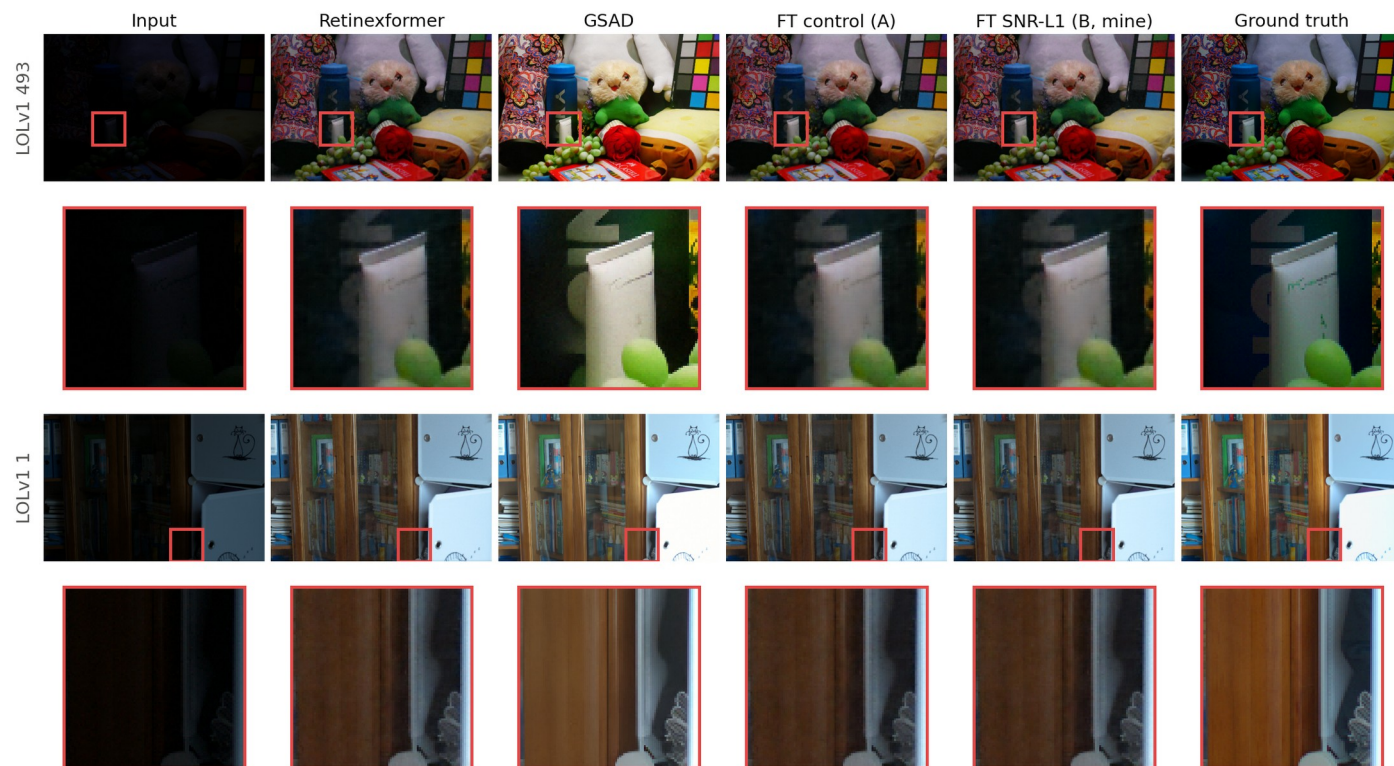
# Efficiency: quality vs. speed (one Tesla T4)



- SCI: 1.5 ms, low quality
- Retinexformer: 151 ms, best PSNR
- LLFormer: 874 ms, 24.5 M params
- GSAD: 2.4–4.9 s per image



# Limitations and failure cases



- Only 15 LOL-v1 test images
- One seed per run; limited free GPU time
- SNR estimate also marks edges as noise
- Detector used as-is (not fine-tuned)
- SNR-Aware: released images only; GSAD on 200 ExDark images

# Plan for the second half

**Weeks 1–2**

SNR guidance inside IG-MSA; noise estimate that ignores edges

**Weeks 3–4**

Sweep  $\alpha$  and weighting; 3 seeds per setting

**Weeks 5–6**

Train best variant from scratch; larger test sets

**Week 7**

Fine-tune YOLOv8 on enhanced vs raw ExDark

**Week 8**

Final report and code release

# Conclusions

- Reproduced Retinexformer (25.15 dB) + 8 baselines under one protocol
- Found: GT-brightness trick (GSAD), LOL-v2-real / LOL-v1 overlap
- Raw images beat every enhancer for YOLOv8m (mAP50 0.662)
- SNR-weighted loss: consistent but negligible gain in dark regions
- Next: noise awareness inside attention, larger tests

Key references: Cai et al., Retinexformer, ICCV 2023 · Xu et al., SNR-Aware, CVPR 2022 · Hou et al., GSAD, NeurIPS 2023 · Wang et al., LLFormer, AAAI 2023 · Guo et al., Zero-DCE, CVPR 2020 · Ma et al., SCI, CVPR 2022 · Li et al., LLIE survey, TPAMI 2022 · Loh & Chan, ExDark, CVIU 2019 (full list in the report)